

UNFOLDED PRIMAL-DUAL ALGORITHM ESTIMATING TIME-VARYING COVID-19 REPRODUCTION NUMBERS

Benjamin Pineau, Barbara Pascal, Sébastien Bourguignon

Nantes Univ., École Centrale Nantes, CNRS LS2N, UMR 6004, F-44000 Nantes, France

benjamin.pineau@eleves.ec-nantes.fr, barbara.pascal@cnrs.fr, sebastien.bourguignon@ec-nantes.fr

Abstract—During an epidemic outbreak, accurate real-time monitoring of the intensity of the pathogen propagation is key to implement efficient countermeasures rapidly enough to avoid dramatic epidemic waves. However, this represents an immense challenge for two main reasons. First, for an emerging disease such as COVID-19, very few prior knowledge is available. Second, in a crisis context, the quality of collected data might be very low. One popular indicator to quantify the pathogen transmissibility, is the time-varying reproduction number, defined as the average number of infections generated by a typical contagious individual if all conditions remained the same. Recently, a variational estimator of the reproduction number robust to low data quality has been proposed, relying on a customized primal-dual minimization algorithm. This work proposes to unfold this sophisticated optimization scheme into a residual convolutional neural network. The obtained network, designed to estimate time-varying reproduction numbers from infection counts, is trained on genuinely crafted synthetic data. It is then compared to state-of-the-art estimators and shown to produce very consistent estimates of the reproduction number when tested on real COVID-19 counts from several countries at different phases of the pandemic. A documented Python toolbox is publicly available to reproduce all presented results.

Index Terms—Primal-dual algorithms, proximal operators, unfolding, convolutional neural networks, reproduction number, COVID-19.

I. INTRODUCTION

Context. The COVID-19 pandemic heavily stroke societies worldwide and brought to the attention of the whole scientific community the challenges raised by the surveillance of an epidemic. Indeed, not only prior knowledge on the propagation mechanisms of an emerging disease is scarce, but also the infection counts collected in a crisis context may be severely corrupted by reporting errors, administrative noise and underestimation during days-off [1]. A popular indicator to quantify in real-time the pathogen transmissibility is the time-varying reproduction number, defined as the number of secondary cases generated by a typical infectious individual [2]. A reference estimator of the reproduction number of viral epidemics has been proposed in [3] and implemented in the widely used EpiEstim software [4]. Yet, EpiEstim is significantly impaired by the unprecedented low quality of COVID-19 data [1], [5], [6]. Recently, advanced inverse problem solving tools have been customized to the epidemiological model proposed in [3] and yielded robust variational estimators of the reproduction number, defined as the minimizers of regularizing objective functions [5], [6]. Yet, the induced computational burden and the need to fine-tune the level of regularization to get accurate estimates severely limit their practical use.

Related works. An innovative strategy to tackle these limitations while keeping a solid connection to the considered model is algorithm unfolding [7]. It consists in reinterpreting iterations of a minimization algorithm as layers of a neural network [8]. Major advantages of unfolded networks are that their architecture is specifically adapted to the model, and that they can encode in a versatile way the regularity brought by priors, which can thus be learned directly from data. Moreover, such model-based architectures generally require

significantly less training data than standard convolutional neural networks and have achieved high performance in very diverse reconstruction problems [8]. This makes unfolded algorithms very well adapted to the present context where a solid epidemiological basis is required and access to training data is a challenge per se. Primal-dual algorithms have been successfully unfolded into neural networks for inverse problems in both image [9], [10] and signal [11] processing. Yet, most works assumed an additive Gaussian noise model, while infection counts are typically modeled as a Poisson process. [2], [3]. Up to the authors' knowledge, the few papers unfolding primal-dual algorithms under a data-dependent Poisson noise model avoid the use of the corresponding proximity operator as an activation function, either by unfolding an Alternating Descent of Multiplier Method scheme [12], or by replacing it with a trainable neural network [9], or even by using a Gaussian approximation [13].

Outline and contributions. The first major contribution of the present work consists in the exact unfolding of the primal-dual Chambolle-Pock scheme [14], particularized to the estimation of the reproduction number under the scaled Poisson model proposed in [6]. Second, a training strategy relying on carefully designed synthetic data is proposed and validated through intensive numerical simulations, covering different variance levels in synthetic counts to mimic unequal levels of intrinsic fluctuations in real epidemiological data. The influence of the structure hyperparameters of the unfolded network is systematically explored to determine the width and depth that achieve the best compromise between estimation accuracy and small number of trainable weights. The proposed unfolded network is then quantitatively compared against state-of-the-art methods on synthetic data. Finally, it is shown to yield very consistent reproduction number estimates when applied on real COVID-19 counts from the publicly available Johns Hopkins University repository.⁵ All experiments are reproducible via a publicly available Python toolbox.⁶ Sec. II introduces the scaled Poisson model [15], describes the reference estimator EpiEstim [4], the variational estimator from [5] and the primal-dual Chambolle-Pock algorithm to compute it. The unfolding of this scheme and the learning strategy based on synthetic data are detailed in Sec. III. Intensive numerical experiments on both synthetic and real data are presented and discussed in Sec. IV. Finally, concluding remarks and perspectives are drawn in Sec. V.

II. EPIDEMIC MODELING AND MONITORING TOOLS

Scaled Poisson model for viral epidemics. Epidemic monitoring largely relies on renewal equations to model the dynamics of infection counts driven by a time-varying reproduction number [2]. A very popular framework in epidemiology, originally proposed in [3], models transmission as a Poisson process. Elaborating on [2], [3], an extended *scaled* Poisson model has been introduced in [15] to specifically account for the large intrinsic variability of infection counts during an outbreak such as the COVID-19 pandemic. The

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Algorithm 1 Chambolle-Pock algorithm for solving (4)

Input: $\mathbf{Z}, \Phi \in \mathbb{R}^T$, $\gamma > 0$, $\lambda > 0$

Parameters: $\rho, \sigma > 0$ s.t. $\rho\sigma\|\mathbf{D}_2\|_{\text{op}}^2 < 1$, $\varepsilon > 0$, $k_{\max} \in \mathbb{N}$

- 1: **Prepare data:** $\tilde{\mathbf{Z}} = \mathbf{Z}/\gamma$, $\tilde{\Phi} = \Phi/\gamma$
 - 2: **Initialize:** $\mathbf{R}^{[0]} = \bar{\mathbf{R}}^{[0]} \in \mathbb{R}_+^T$, $\mathbf{U}^{[0]} \in \mathbb{R}^{T-2}$, $k = 0$, $\Psi = \varepsilon + 1$
 - 3: **while** $k < k_{\max}$ **and** $\Psi > \varepsilon$ **do**
 - 4: $\mathbf{U}^{[k+1]} = \text{prox}_{\sigma(\lambda\|\cdot\|_1)^*}(\mathbf{U}^{[k]} + \sigma\mathbf{D}_2\bar{\mathbf{R}}^{[k]})$ # dual
 - 5: $\mathbf{R}^{[k+1]} = \text{prox}_{\rho\mathbf{D}_{\text{KL}}(\tilde{\mathbf{Z}}|\cdot\odot\tilde{\Phi})}(\mathbf{R}^{[k]} - \rho\mathbf{D}_2^\top\mathbf{U}^{[k+1]})$ # primal
 - 6: $\bar{\mathbf{R}}^{[k+1]} = 2\mathbf{R}^{[k+1]} - \mathbf{R}^{[k]}$ # inertial
 - 7: $k = k + 1$, $\Psi = \|\mathbf{R}^{[k]} - \mathbf{R}^{[k-1]}\|_2$
 - 8: **end while**
 - 9: **Output:** $\hat{\mathbf{R}}_\lambda^{\text{Pen-KL}} = \mathbf{R}^{[k]}$
-

infection count at day t , denoted by Z_t , conditionally to past counts $Z_1, \dots, Z_{t-1}$, is assumed proportional to a Poisson random variable:

$$\left(\frac{Z_t}{\gamma} \mid Z_1, \dots, Z_{t-1}, R_t\right) \sim \mathcal{P}\left(\frac{R_t\Phi_t}{\gamma}\right) \quad (1)$$

where $\gamma > 0$ modulates the counts variance, $R_t \geq 0$ is the time-varying reproduction number at time t and Φ_t denotes the global infectiousness, defined as the weighted sum of past counts $\Phi_t = \sum_{v=1}^{\tau_\Phi} \phi(v)Z_{t-v}$. The weights account for the random delay between primary and secondary infections, encoded in the serial interval function ϕ [2], [3], and τ_Φ reflects the maximum duration of contagiousness, which is a typical time scale of the epidemic. For viral epidemics, ϕ is often well approximated by a gamma distribution, which, for COVID-19, has a mean (resp. standard deviation) of 6.6 (resp. 3.5) days and a support of $\tau_\Phi = 25$ days [16]. The model in [2], [3] corresponds to Model (1) with $\gamma = 1$.

Reference reproduction number estimators. The maximum likelihood estimator under Model (1) with $\gamma = 1$ has been derived in [2]:

$$\hat{R}_t^{\text{ML}} = \frac{Z_t}{\Phi_t}, \quad (2)$$

which extends directly to any value of γ . Yet, raw COVID-19 infection counts, in gray in top plots of Fig. 6, are severely corrupted by reporting errors and administrative noise [1]. The maximum likelihood estimates, in gray in bottom plots of Fig. 6, thus exhibit erratic variations at a time scale far smaller than τ_Φ . These fluctuations, induced by noise, harm the reliability of the ML estimates. A more robust smoothed estimator has been introduced in [3]:

$$\hat{R}_{t,\tau}^{\text{EpiEstim}} = \frac{\sum_{v=t-\tau+1}^t Z_v + a}{\sum_{v=t-\tau+1}^t \Phi_v + 1/b} \quad (3)$$

with $a = 1$ and $b = 5$. The smoothing window length τ has been found to be optimal for several viral epidemics when set to $\tau = 7$ days [3]. The estimator (3), implemented in the publicly available EpiEstim R package [4], is considered as state-of-the-art in epidemiology [1]. Although more regular, the EpiEstim reproduction number estimates from real COVID-19 infection counts, drawn in green in Fig. 6 (bottom), still suffer from the low quality of data.

Regularized variational estimates. To improve robustness to the low quality of real data, variational estimators have recently been designed in an inverse-problem framework [5], [6]. Given a T -day-long time series of counts $\mathbf{Z} = (Z_1, \dots, Z_T) \in \mathbb{N}^T$, yielding the infectiousness time series $\Phi = (\Phi_1, \dots, \Phi_T) \in \mathbb{R}_+^T$, the proposed regularized estimator of $\mathbf{R} = (R_1, \dots, R_T)$ is defined as the minimizer of a functional balancing the negative log-likelihood of Model (1) and an

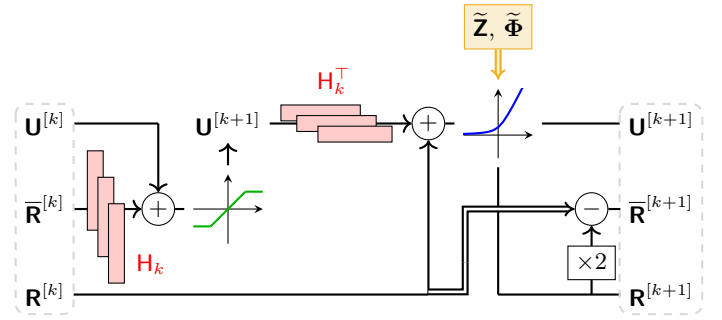


Fig. 1: **Layer k of the Unfolded Chambolle-Pock algorithm.** Trainable convolutional operators are represented in red, the activation functions from the data-fidelity term (resp. the penalization) in blue (resp. green) and the residual connection with a double line.

ℓ_1 -norm-based penalization favoring piecewise linearity:

$$\hat{\mathbf{R}}_\lambda^{\text{Pen-KL}} = \underset{\mathbf{R} \in \mathbb{R}_+^T}{\text{argmin}} \sum_{t=1}^T \mathbf{d}_{\text{KL}}\left(\frac{Z_t}{\gamma} \mid \frac{R_t\Phi_t}{\gamma}\right) + \lambda\|\mathbf{D}_2\mathbf{R}\|_1 \quad (4)$$

where $\mathbf{d}_{\text{KL}}(\mathbf{Z}|\mathbf{P}) = \mathbf{Z} \ln(\mathbf{Z}/\mathbf{P}) + \mathbf{P} - \mathbf{Z}$, if $\mathbf{Z} > 0$, $\mathbf{P} > 0$; $\mathbf{P}$, if $\mathbf{Z} = 0$, $\mathbf{P} \geq 0$; ∞ , otherwise, is the Kullback-Leibler divergence, $\mathbf{D}_2$ is the discrete Laplacian operator, defined by $(\mathbf{D}_2\mathbf{R})_t = (R_{t+2} - 2R_{t+1} + R_t)/4$ for all $t \in \{1, \dots, T-2\}$, and the regularization parameter $\lambda > 0$ controls the level of temporal regularity of the estimate. Following [5], [6], the scale parameter is set according to the heuristic $\gamma \propto \text{std}(\mathbf{Z})$. Because of the ℓ_1 -norm, the objective function in (4) is nonsmooth, hence its minimization requires to resort to specific techniques, e.g., proximal algorithms [17]. Because the gradient of $\mathbf{d}_{\text{KL}}$ is not Lipschitzian and the ℓ_1 -norm is composed with the linear operator $\mathbf{D}_2$, primal algorithms are not adapted. It has been proposed in [5], [6] to minimize (4) using the primal-dual Chambolle-Pock scheme [14] sketched in Alg. 1, where $\odot$ denotes the component-wise multiplication and $\mathbf{D}_{\text{KL}}$ denotes the sum of element-wise Kullback-Leibler divergences. Alg. 1 alternates dual (step 4), primal (step 5) and inertial (step 6) updates. The primal and dual updates consist in linear operations followed by the application of proximal operators, which are nonlinear functions acting component-wise, as both $\mathbf{D}_{\text{KL}}$ and $\|\cdot\|_1$ are separable. The closed-form expressions of $\text{prox}_{\sigma(\lambda\|\cdot\|_1)^*}(\cdot)$ and $\text{prox}_{\rho\mathbf{D}_{\text{KL}}(\tilde{\mathbf{Z}}|\cdot\odot\tilde{\Phi})}(\cdot)$ can be found, e.g., in [6] and their component-wise behavior is sketched in Fig. 1 in green and blue respectively. Usually, the convergence condition is saturated by choosing descent steps $\rho = \sigma = 0.99/\|\mathbf{D}_2\|_{\text{op}}$, where $\|\cdot\|_{\text{op}}$ denotes the spectral norm. A major practical bottleneck for the use of the penalized Kullback-Leibler estimator (4) is that the accuracy of the estimate strongly depends on the precise fine-tuning of the regularization parameter λ [5], [6]. Very recently, a strategy for data-driven selection of an optimal parameter $\lambda^\dagger$ has been proposed [15], based on the minimization of an original autoregressive Poisson unbiased risk estimate specifically designed for Model (1). Plugging $\lambda^\dagger$ into (4) yields the APURE estimator $\hat{\mathbf{R}}^{\text{APURE}} = \hat{\mathbf{R}}_{\lambda^\dagger}^{\text{Pen-KL}}$.

III. UNFOLDED CHAMBOLLE-POCK NEURAL NETWORK

The original idea of algorithm unfolding [7] was motivated by the structural similarity between proximal iterative algorithms and neural networks, which both alternate linear operations and nonlinear activation functions. Unfolding an algorithm consists in reinterpreting its K first iterations as K layers of a neural network. Some elements of the iterative algorithm, typically linear operators, such as $\mathbf{D}_2$ in Alg. 1, are interpreted as trainable operators and possibly untied across layers into a sequence of operators. In this setting, the

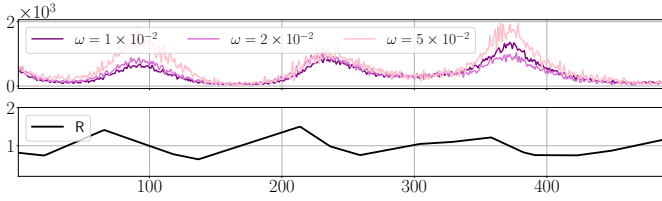


Fig. 2: **Annotated synthetic epidemiological data.** *Top*: synthetic infection counts under Model (1) with $\gamma = \omega \times \delta$ for different variance levels ω ; *bottom*: underlying realistic synthetic reproduction number.

activation functions are not chosen arbitrarily but instead are guided by the problem at hand [18], e.g., in Alg. 1 they correspond to the proximity operators of the two terms of the functional (4). Inertial steps, e.g., line 6 of Alg. 1, are encoded as residual connections [18]. **Proposed unfolded architecture.** In this work, Alg. 1 is unfolded into a neural network specifically designed to estimate the time-varying reproduction number from infection counts. To that aim, the K first iterations of Alg. 1 are interpreted as composite convolutional layers with residual connections, sketched in Fig. 1, writing:

$$\begin{cases} \mathbf{U}^{[k+1]} &= \text{prox}_{\sigma(\lambda \|\cdot\|_1)^*} \left(\mathbf{U}^{[k]} + \sigma \mathbf{H}_k \bar{\mathbf{R}}^{[k]} \right) \\ \mathbf{R}^{[k+1]} &= \text{prox}_{\rho \mathbf{D}_{\text{KL}}(\tilde{\mathbf{Z}} \cdot \odot \tilde{\Phi})} \left(\mathbf{R}^{[k]} - \rho \mathbf{H}_k^\top \mathbf{U}^{[k+1]} \right) \\ \bar{\mathbf{R}}^{[k+1]} &= 2\mathbf{R}^{[k+1]} - \mathbf{R}^{[k]} \end{cases} \quad (5)$$

where the linear operator $\mathbf{D}_2$ has been untied into a sequence of trainable operators $\{\mathbf{H}_k, k = 1, \dots, K\}$. The convolutional neural network framework is adapted to the epidemiological model of Sec. II by considering that the operator $\mathbf{H}_k$ consists in a set of convolutions, computed in valid mode, with J independent trainable filters of length τ_Φ , the typical time scale of the COVID-19 pandemic. The activation functions are preserved as those encoding the prior knowledge and the model presented in Sec. II, and respectively correspond to the proximity operators of the convex conjugate of the ℓ_1 -norm, exactly coinciding with the HardTanh function [6] sketched in green in Fig. 1, and of the Kullback-Leibler divergence, drawn in blue, which explicitly depends on the scaled infection counts and infectiousness data.¹ It is worth noting that the proximity operator of $\mathbf{d}_{\text{KL}}$ constitutes an original activation function, although it resembles the bent identity. In all papers found by the authors addressing algorithm unfolding with Poisson inverse problems, the use of $\mathbf{d}_{\text{KL}}$ as a data-fidelity term, and hence of its proximity operator as an activation function, is circumvented. For example in [13] $\mathbf{d}_{\text{KL}}$ is replaced by a second-order Taylor series approximation, while in [9] its proximity operator is replaced by a trainable convolutional neural network. A major originality of the proposed unfolded network thus consists in using the proximity operator of the Kullback-Leibler divergence, exactly suited to Poisson models, as a nonstandard activation function. The unfolded Chambolle-Pock network is finally built as the concatenation of K composite layers of type (5), applied to constant input vectors: $\mathbf{R}_t^{[0]} = \bar{\mathbf{R}}_t^{[0]} = 1$ and $\mathbf{U}_t^{[0]} = 0 \forall t$. The last layer only outputs the primal variable $\mathbf{R}^{[K]}$. The regularization parameter is arbitrarily fixed at $\lambda = 1$, as both the magnitude of the trainable operators and the finite number of layers [11], [19] control the regularization level. All remaining parameters are kept unchanged compared to the setting described in Sec. II: $\rho = \sigma = 0.99/\|\mathbf{D}_2\|_{\text{op}}$.

Learning untied linear operators. Denoting by $\mathbf{h}_{k,j} \in \mathbb{R}^{\tau_\Phi}$ the j^{th} filter of $\mathbf{H}_k$, the linear operator at layer k , the proposed unfolded neural network has $K \times J \times \tau_\Phi$ trainable weights, stored in

¹From a deep learning perspective, using scaled data $\tilde{\mathbf{Z}} = \mathbf{Z}/\gamma$ and $\tilde{\Phi} = \Phi/\gamma$ can be interpreted as data normalization which ease stable learning.

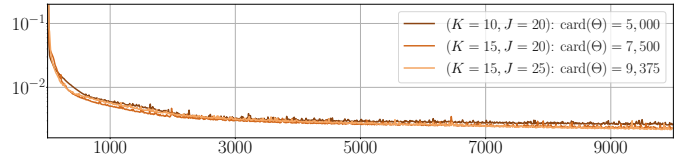


Fig. 3: **Convergence of the training procedure.** Loss (6) along epochs while training unfolded networks with different numbers of layers K and filters J on the multiple variance level dataset $\mathcal{S}_\Omega$.

$\Theta = \{\mathbf{h}_{k,j}, k = 1, \dots, K, j = 1, \dots, J\}$. Training this network to estimate reproduction numbers consists in inferring, from a dataset $\mathcal{S} = \{(\mathbf{R}_{[s]}^*, \mathbf{Z}_{[s]}), s = 1, \dots, S\}$ composed of pairs of ground truth reproduction numbers $\mathbf{R}_{[s]}^* \in \mathbb{R}_+^T$ and associated infection counts $\mathbf{Z}_{[s]} \in \mathbb{R}_+^T$, the weights Θ that minimize the discrepancy between the network output and the ground truth. This discrepancy is quantified by the standard squared ℓ_2 loss defined as:

$$\mathcal{L}(\Theta; \mathcal{S}) = \frac{1}{S \times T} \sum_{s=1}^S \|\mathbf{N}_\Theta(\mathbf{Z}_{[s]}) - \mathbf{R}_{[s]}^*\|_2^2 \quad (6)$$

where $\mathbf{N}_\Theta(\mathbf{Z}_{[s]})$ denotes the unfolded Chambolle-Pock network with weights Θ and in which the proximity operator of $\mathbf{D}_{\text{KL}}$ is fed with counts $\tilde{\mathbf{Z}}_{[s]} = \mathbf{Z}_{[s]}/\gamma$ and infectiousness $\tilde{\Phi}_{[s]} = \Phi_{[s]}/\gamma$. Untied linear operators, stored in Θ , are then learned by minimizing the training loss (6) computed over $\mathcal{S}$:

$$\Theta^{\mathcal{S}} \in \underset{\Theta \in \mathbb{R}^{K \times J \times \tau_\Phi}}{\text{Argmin}} \mathcal{L}(\Theta; \mathcal{S}). \quad (7)$$

Finally, for a time series $\mathbf{Z} \in \mathbb{N}^T$, the Unfolded Chambolle-Pock estimate is defined as the output of the network trained on the dataset $\mathcal{S}$: $\hat{\mathbf{R}}^{\mathcal{S}\text{-UnCP}} = \mathbf{N}_{\Theta^{\mathcal{S}}}(\mathbf{Z})$ and referred to as $\mathcal{S}$ -UnCP in the following.

Synthetic datasets. To circumvent the lack of abundant annotated epidemiological data, the second major contribution of this work consists in training the network on carefully designed synthetic data. The major challenge consists in designing realistic time-varying reproduction numbers to be used as ground truths. To that aim, the present work relies on the procedure originally proposed in [20], which provides some synthetic ground truths $\mathbf{R}^*$ accompanied by reference scales $\delta > 0$. While $\mathbf{R}^*$ encodes the epidemic dynamics, δ independently models the typical range of counts. Varying δ enables to mimic different population sizes. Then, the scale parameter is decomposed into $\gamma = \omega \times \delta$, so that δ is modulated by a variance level $\omega > 0$ which is invariant to both the propagation scenario encoded in $\mathbf{R}^*$ and to the reference scale δ . Finally, $\mathbf{R}^*$ is injected into Model (1) with $\gamma = \omega \times \delta$ to generate synthetic infection counts $\mathbf{Z}$ at variance level ω . An example of synthetic $\mathbf{R}^*$ is drawn in the bottom plot of Fig. 2, with three examples of associated synthetic $\mathbf{Z}$ of low to high variance levels ω , in top plot, which appear realistic compared to real COVID-19 counts displayed in top plots of Fig. 6.

Performance evaluation on synthetic data. The performance of an estimate $\hat{\mathbf{R}}$ is quantified through two complementary indicators: first the signal-to-noise ratio in decibels measuring accuracy with respect to ground truth: $\text{SNR} = 10 \times \log_{10} \left(\|\mathbf{R}^*\|_2^2 / \|\hat{\mathbf{R}} - \mathbf{R}^*\|_2^2 \right)$; second the squared ℓ_2 -norm of the first-order derivative of $\hat{\mathbf{R}}$: $\|\mathbf{D}_1 \hat{\mathbf{R}}\|_2^2 = \sum_{t=1}^{T-1} (\hat{\mathbf{R}}_{t+1} - \hat{\mathbf{R}}_t)^2$, is a smoothness measure which, when too large, strongly suggests erratic fluctuations induced by administrative noise.

IV. NUMERICAL EXPERIMENTS

Training and test datasets. First, $L = 10$ independent training datasets $\mathcal{S}_{\omega^{(\ell)}}$, $\ell = 1, \dots, L$ at different variance levels $\omega^{(\ell)}$, $\ell = 1, \dots, L$, logarithmically spaced between 0.01 and 0.05, are con-

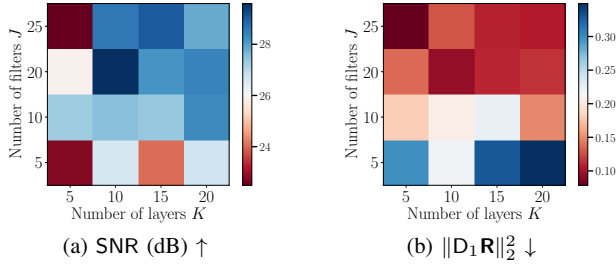


Fig. 4: **Performance depending on network width and depth.** The number of layers and trainable filters per layer are varied. The networks are trained on a synthetic dataset composed of 300 examples at fixed variance level $\omega^{(5)} \simeq 2 \times 10^{-2}$. *Left*: Estimation accuracy quantified by the signal to noise ratio; *right*: smoothness of the estimates measured by the squared norm of the first order derivative.

structured. To that aim, for each variance level $\omega^{(\ell)}$, C independent ground truth reproduction numbers² of duration $T = 490$ days $\mathbf{R}_{[c]}^*$, $c = 1, \dots, C$ are generated following [20], accompanied by reference scales δ_c , $c = 1, \dots, C$. For each of the C ground truths, Q synthetic infection counts realizations $\mathbf{Z}_{[c,q]}$, $q = 1, \dots, Q$ are drawn³ from Model (1) with scale parameters $\gamma_c = \omega^{(\ell)} \times \delta_c$ using the code⁴ accompanying [21], resulting in a dataset $\mathcal{S}_{\omega^{(\ell)}}$ of $S = C \times Q$ pairs. Finally, these datasets are merged into a multiple variance level dataset $\mathcal{S}_{\Omega} = \bigcup_{\ell=1}^L \mathcal{S}_{\omega^{(\ell)}}$. Independently, $L = 10$ similar test datasets at same variance levels are generated following the same procedure. In practice, $C_{\text{train}} = 60$ and $C_{\text{test}} = 4$ ground truths are considered and, for each of them, $Q_{\text{train}} = 5$ and $Q_{\text{test}} = 200$ infection counts realizations are drawn. The multiple variance level training dataset thus contains $L \times C_{\text{train}} \times Q_{\text{train}} = 3,000$ pairs. A synthetic $\mathbf{R}^*$ and associated $\mathbf{Z}$ for different $\omega^{(\ell)}$ are drawn in Fig. 2.

Learning setup. The unfolded network is trained by solving (7) for each of the L training datasets at fixed variance level $\mathcal{S}_{\omega^{(\ell)}}$ and for the multiple variance level dataset $\mathcal{S}_{\Omega}$ described above. The filters are initialized at $[1, 2, 1]$ padded with zeros to reach a length of $\tau_{\Phi} = 25$, then the ADAM optimizer is run for a fixed number of 10,000 epochs with mini-batches of size 60 and a constant learning rate of $\text{lr} = 10^{-4}$ to avoid fast convergence toward a poor local minimum observed with the default ADAM learning rate of 10^{-3} . The evolution of the loss (6) along epochs while training on the multiple variance level dataset $\mathcal{S}_{\Omega}$, displayed in Fig. 3, shows clear stabilization for different numbers of layers and filters. The Unfolded Chambolle-Pock network is implemented using PyTorch (ver. 2.9.1) under Python (ver. 3.12.6). The GPU-based code is run on a server equipped with a single NVIDIA Tesla T4 GPU (16 GB of GPU memory), using NVIDIA driver (ver. 550.144.03) with CUDA support (up to ver. 12.4).

Comparison of architectures. The first set of experiments aims at fine-tuning the structure hyperparameters of the unfolded algorithm, namely the number K of layers and the number J of filters per layer. To that aim, K is varied in $\{5, 10, 15, 20\}$ and J in $\{5, 10, 20, 25\}$, inspired by the typical ranges explored in algorithm unfolding, e.g., in [10], and the different resulting architectures are trained on the medium variance dataset $\mathcal{S}_{\omega^{(5)}}$. Fig. 4 displays the average signal-to-noise ratio and smoothness achieved on the associated test dataset at variance level $\omega^{(5)}$, as a function of K and J . For small numbers of layers or filters, one can observe in Fig. 4a that the

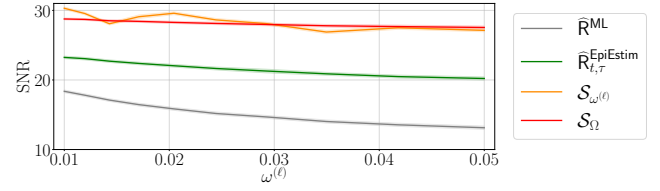


Fig. 5: **Compared estimation performance.** Average SNR and 95% Gaussian confidence intervals computed on the $L = 10$ test datasets with variance levels $\omega^{(\ell)}$, $\ell = 1, \dots, L$ for ML (gray), EpiEstim (green), $\mathcal{S}_{\omega^{(\ell)}}\text{-UnCP}$ (orange) and $\mathcal{S}_{\Omega}\text{-UnCP}$ (red) estimators.

SNR is low, indicating poor generalization capacity of the unfolded network. Conversely, as soon as K (resp. J) is larger than 10 (resp. 20), the performance of the $\mathcal{S}_{\omega^{(5)}}\text{-UnCP}$ estimator stabilizes at its highest level. Furthermore, Fig. 4b shows that in this range of hyperparameters the obtained estimates are smooth, attesting the ability of the trained unfolded network to remove erratic fluctuations induced by noise. To get the best performing network at the lowest inference and memory costs, the number of layers and filters are set to $K = 10$ and $J = 20$, for a total of 5,000 trainable weights.

Estimation performance. The second set of experiments aims at assessing the estimation performance of the unfolded network when trained, either on datasets with specific variance level, or on the multiple variance levels dataset, respectively yielding the $\mathcal{S}_{\omega^{(\ell)}}\text{-UnCP}$ and $\mathcal{S}_{\Omega}\text{-UnCP}$ estimators, and to compare them to the naive ML (2) and the reference EpiEstim (3) estimators. Fig. 5 reports the SNR of the different estimates on each of the L test datasets as a function of the variance level. The proposed unfolded network, whatever the training dataset considered, significantly outperforms both ML and EpiEstim. The estimation performance of both $\mathcal{S}_{\omega^{(\ell)}}\text{-UnCP}$ and $\mathcal{S}_{\Omega}\text{-UnCP}$ predictably decreases as the variance level increases, yet this decrease is very slow and the performance remains far better than those of the reference EpiEstim estimator, even at high variance level. Finally, the strategy consisting in training on the multiple variance level dataset $\mathcal{S}_{\Omega}$ does not induce any performance loss compared to training on a dataset at the exact same variance level as the test dataset. This is a very important observation as it enables the training to be blind to the variance level, yielding a robust estimate $\mathcal{S}_{\Omega}\text{-UnCP}$ automatically adapting to the data. All estimators have very low empirical variance across the test datasets, leading to almost invisible 95% Gaussian confidence intervals in Fig. 5.

Estimation of the COVID-19 reproduction number from real data. The third set of experiments aims at demonstrating the ability of the proposed unfolded network to estimate reproduction numbers from real epidemiological counts, namely the COVID-19 infection counts in several countries and at different 490-day periods. Real COVID-19 counts, displayed in gray in top plots of Fig. 6, are downloaded from the Johns Hopkins repository.⁵ They not only exhibit large intrinsic variance but are also corrupted by severe administrative noise, which dramatically harms accurate estimation of the reproduction number. To circumvent this issue, counts are preprocessed with the median filtering described in detail in [20, Eq. (4)] with a 15-day window and $\alpha = 0.1$, yielding the corrected counts displayed in black. As for synthetic data, the estimates from ML and EpiEstim, displayed in gray and green respectively in bottom plots of Fig. 6, suffer from rapid fluctuations, unrealistic compared to the time scale of COVID-19 pandemic of $\tau_{\Phi} = 25$ days. On the contrary, the proposed $\mathcal{S}_{\Omega}\text{-UnCP}$ estimator, displayed in red, yields a smooth estimate that is far more convincing. For exhaustive com-

²For the sake of compactness, once the variance level $\omega^{(\ell)}$ fixed, the superscript ℓ is implicit in $\mathbf{R}^*$, δ and $\mathbf{Z}$.

³In practice, to avoid extinction of the synthetic epidemic propagation, the samples in which three successive counts vanish are discarded.

⁴See create_data.py in github.com/elasalle/EpiJointSpatiotempEstim/

⁵coronavirus.jhu.edu/

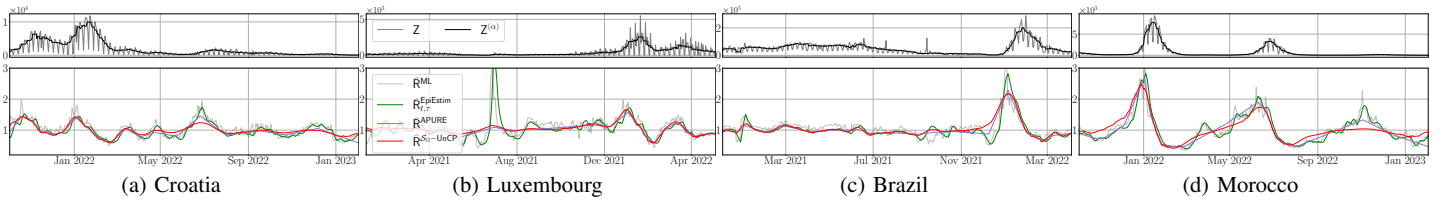


Fig. 6: **Estimation of the reproduction number from real COVID-19 counts.** *Top:* Raw infection counts from Johns Hopkins University (gray) and pre-processed counts through median filtering (black). *Bottom:* Reproduction number estimates from ML (gray), EpiEstim [3], [4] (green), APURE (purple) [15], and the proposed unfolded Chambolle-Pock network trained on multiple variance levels S_{Ω} -UnCP (red).

	Croatia	Luxembourg	Brazil	Morocco
ML [2]	3.648	13.986	4.288	5.891
EpiEstim [3], [4]	0.373	2.052	0.760	0.899
APURE [15]	0.099	0.065	0.161	0.419
S_{Ω} -UnCP	0.083	0.109	0.181	0.412

TABLE I: **Regularity of reproduction number estimates.** Squared ℓ_2 -norm of the first-order derivative of reproduction number estimated from real COVID-19 counts using state-of-the-art methods vs. the proposed unfolded Chambolle-Pock neural network. $\hat{R}^{S_{\Omega}\text{-UnCP}}$

parison, the APURE estimator [15] is displayed in purple. It appears to provide similar quality estimates as the S_{Ω} -UnCP estimator. This qualitative comparison is complemented by a quantitative analysis of the smoothness of estimated reproduction numbers reported in Tab. I which confirms that APURE and S_{Ω} -UnCP yield far smoother estimates. Yet, APURE requires five hours of computations to yield a $T = 490$ -day-long estimate on a 2 GHz quad-core Intel Core i5 processor, while inference of S_{Ω} -UnCP takes less than one second.

V. CONCLUSION AND PERSPECTIVES

The Chambolle-Pock primal-dual algorithm [14] has been unfolded into a convolutional neural network with a residual connection specifically designed to estimate the reproduction number of an epidemic from collected infection counts. This network has then been trained on carefully crafted synthetic data and its estimation performance at fixed variance level assessed through intensive numerical experiments. Furthermore, a multiple variance level training strategy has been proposed and shown to yield a very accurate estimator which is blind to the variance level, unknown in practice, and hence well-suited to monitor a real epidemic. Finally, the unfolded network, although trained only on synthetic data, has been shown to provide very consistent estimates when applied on severely corrupted COVID-19 case counts. Python codes reproducing all the experiments are publicly available.⁶ This work will be complemented by a systematic study of the robustness of the network to perturbation of its inputs, quantified by means of Lipschitz constants [10], which is key to enable broader use of the proposed estimator in critical context, such as epidemic outbreaks. Beyond robustness, epidemiological indicators need to be accompanied by a confidence level to be used by decision makers. To that aim, the unsupervised conformal prediction framework adapted to Poisson models proposed in [22] will be combined with the autoregressive Poisson unbiased risk estimate proposed in [15] to yield confidence regions for the reproduction number estimates obtained with the proposed unfolded network.

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⁶github.com/BenjaminPINEAU/Covid-R-Unroll